PROJECT SUTRA — Visual Role Infographics

TEAM OFFGRID SIMPLE STEP-BY-STEP SUBSYSTEM GUIDE

TEAM OFFGRID
Finals: Sept 3, 2026

🛸 Subsystem A: Autonomous Navigation & Flight Control

ASSIGNED TO: ROHITH KUMAR

1. PX4 Offboard Mode — Micro-XRCE-DDS 8888 → **2. TrajectorySetpoint** — 10m Takeoff & Square Waypoint → **3. ORCA Avoidance** — Prevents Drone Collision → **4. Gazebo Flight** — Clean 60 FPS Drone Motion

- ✔ **What Rohith Builds:** C++ ROS 2 node `px4_offboard_node.cpp`.
- ✔ **Day 1 Goal:** Install PX4 SITL & verify DDS connection.
- ✔ **How it Works:** Sends 3D velocity vectors to PX4 flight controller.
- ✔ **Success Metric:** Drone flies square circuit with error < 1.0m.

🌐 Subsystem B: Swarm Comms Mesh & Deep JSCC Neural Encoder

ASSIGNED TO: NIKHIL (TECH ARCHITECT)

1. 802.11s Mesh Socket — UDP + 868MHz LoRa Link → **2. Deep JSCC Encoder** — Neural Image Compression → **3. RAFT Consensus** — Instant Swarm Leader Failover → **4. Zero Drop Link** — 99.4% Delivery under 60% Loss

- ✔ **What Nikhil Builds:** Python `mesh_socket.py` & Deep JSCC neural model.
- ✔ **Day 1 Goal:** Init repository, configure branches & ROS 2 bridge.
- ✔ **How it Works:** Connects all drones in a mesh network; auto-re-elects leader.
- ✔ **Success Metric:** Zero telemetry loss under 60% simulated RF drop.

👁️ Subsystem C: Tri-Modal AI Perception & Target Geolocation

ASSIGNED TO: VEDANTH SAI RAM

1. Camera Image Feed — RGB + Thermal IR Streams → **2. YOLOv8 TensorRT** — 30+ FPS Edge Target Detection → **3. WGS84 GPS Raycast** — Raycast Centroid to Elevation → **4. GPS Victim Pin** — Precise Lat/Lon Output

- ✔ **What Vedanth Builds:** Python `yolo_tensorrt_node.py` ROS 2 node.
- ✔ **Day 1 Goal:** Set up PyTorch + CUDA + TensorRT environment.
- ✔ **How it Works:** Detects disaster victims in camera feeds and outputs GPS pins.
- ✔ **Success Metric:** Detection precision > 85% at 30+ FPS.

02 Visual Role Infographics (Siva Kesava & Harika)

🗺️ Subsystem D: 3D GIS Dashboard & Ground Control Station

ASSIGNED TO: SIVA KESAVA

1. ROS 2 WebSocket
rosbridge_server Stream

2. Mapbox GL JS 3D
Real-time Satellite Elevation

3. WebGPU Artificial Horizon
Pitch / Roll Telemetry HUD

4. Emergency RTL
One-Click Land Trigger

- ✔ **What Siva Builds:** React component `Mapbox3D.jsx` & HUD widgets.
- ✔ **Day 1 Goal:** Create React app & configure Mapbox token.
- ✔ **How it Works:** Displays live 3D drone position markers on satellite map.
- ✔ **Success Metric:** Zero-latency telemetry rendering on MacBook UI.

📝 Subsystem E: Research, Technical Documentation & Audit

ASSIGNED TO: HARIKA (PMO LEAD)

1. Flight Test Logs
Record Sim Performance

2. Gate Audits (G1-G6)
Verify Metrics & Errors

3. SUTRA Encyclopedia
System Architecture Docs

4. Pitch Deck & Video
Finals Presentation Ready

- ✔ **What Harika Builds:** Master Documentation & Gate Audit Reports.
- ✔ **Day 1 Goal:** Init `docs/SUTRA_Encyclopedia.md` & Gate Templates.
- ✔ **How it Works:** Verifies test results and creates presentation materials for judges.
- ✔ **Success Metric:** 100% complete documentation & audit reports.

03 Day-by-Day Sprint Timeline (Days 1–7 Overview)

DAY 1	DAY 2	DAY 3	DAY 4	DAY 5	DAY 6	DAY 7
Repo & Toolchain Init Baseline	AI Node Build Unit Testing	Gazebo Sim 8 PR v0.1-alpha	Pairwise Sync Flight + Vision	Master Run 5-Subsystem	Stress Test 60% RF Loss	Freeze & Lock 2-Min Video